

Hook Bazaar

Decentralized Marketplace for Uniswap v4 Hooks

Building the Infrastructure Layer for the
Uniswap v4 Hooks Economy

The Problem: 9 Critical Market Failures

Market Infrastructure

- No marketplace for hooks
- Supply & demand exist — no connection
- No competition mechanism

Developer Pain Points

- No monetization model
- No IP protection (code is public)
- No reputation system

Protocol Barriers

- Custom hooks: \$10k–\$100k+
- Development time: weeks–months
- High audit costs & risk

Ecosystem Gaps

- No standardization
- No multi-hook composition
- Unsustainable economics

The Solution: Hook Bazaar

A Decentralized Marketplace & Infrastructure Layer

For Developers

- › Direct monetization
- › IP protection via FHE
- › Reputation profiles
- › Professional exposure

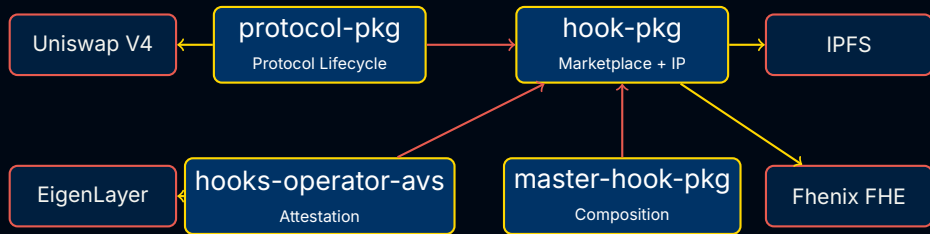
For Protocols

- › Instant deployment
- › **\$100s** vs \$10k+
- › Pre-audited hooks
- › Multi-hook composition

For Ecosystem

- › Self-sustaining model
- › Quality via competition
- › Network effects
- › V4 adoption boost

Architecture & Approach



Core Packages

- **protocol-pkg**: Pool administration
- **hook-pkg**: Development & marketplace

Sponsor Integrations

- **Fhenix CoFHE**: Encrypted hooks for IP protection
- **EigenLayer AVS**: Cryptoeconomic guarantees via

Key Feature: Mathematical Hook Specifications

Hook Specification Format (HSF)

Why It Matters

- › Objective behavior verification
- › Formal state machine definitions
- › Machine-readable specifications
- › Foundation for AVS attestation

What It Enables

- › **Trust:** Verify before integrating
- › **Comparison:** Objective evaluation

```
// Hook State Model
States: {Inactive, Active}
Events: {beforeSwap, afterSwap}
Invariants:
  fee ≤ MAX_FEE
  balance ≥ 0

// Transition
beforeSwap(params):
  require(state == Active)
  fee = computeFee(params)
  emit FeeUpdated(fee)
```

Specifications stored on IPFS

Verified by Eigenlayer operators

Key Feature: EigenLayer AVS Attestation

Cryptoeconomic Guarantees

Why It Matters

- Staked operators verify hook behavior
- Economic penalties for false attestations
- Decentralized trust — no central authority
- Continuous compliance monitoring

Slashing Mechanism

Hook Bazaar ➤ **50% slash:** False positive attestation

